

Transform Validate And Migration Approach(INT to PROD)

Referrals and Appointments

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NOTE- WORK IN PROGRESS

1 Migration Strategy

The proposed migration approach treats the existing INT environment as the primary migration and validation environment. Rather than using `targets.json` as the source of migration data, the migration will leverage the existing curated operational data stored within the legacy INT data structures. This data already contains the business information required to construct Endpoint Catalogue resources, including Service Identifiers, Provider Organisations, Endpoint relationships, Product Identifiers, and routing information.

The objective of the migration is to transform the legacy data model into the new Endpoint Catalogue FHIR R4 compliant structure and validate the resulting resources before any data is promoted to the PROD environment. By performing all transformation and validation activities within INT, the migration process reduces production risk and ensures only high-quality, validated data is deployed. This approach separates data transformation from production deployment, allowing defects, data quality issues, and mapping discrepancies to be identified and resolved before a production release is considered.

Once the transformed resources have successfully passed all validation checks, the validated dataset becomes the approved migration artefact. That approved dataset is then promoted to PROD using a controlled deployment process without requiring any further data transformation activities.

2 FHIR R4 Transformation Process

The migration process begins by extracting data from the legacy INT data sources, including Healthcare Services, Endpoints, Endpoint Templates, and Organisations. These datasets are used to construct the equivalent FHIR R4 resources required by the Endpoint Catalogue.

During transformation, the migration process converts the legacy records into three key resource types:

- Endpoint Templates
- Endpoints
- HealthcareServices

Each resource is populated using information obtained from the existing INT data store and mapped to the FHIR R4 structure expected by the Endpoint Catalogue. This includes generating the necessary identifiers, organisation references, endpoint relationships, provider associations, and metadata required for successful operation within the catalogue.

The transformation process will not automatically migrate every record that exists in the legacy system. Instead, every resource is evaluated against the EPC data model to ensure all mandatory attributes are present and valid. This prevents incomplete or invalid data from entering the new platform and establishes a clean, supportable baseline for the Endpoint Catalogue.

Metric	Count
Source Endpoint Templates	13
FHIR Endpoint Templates Created	13
Source Endpoints	13
FHIR Endpoints Created	13
Source HealthcareServices	10,284
FHIR HealthcareServices Created	10,280
Transformation Failures	4
Transformation Status	PASS

2.1 Transformation Failure Reporting

Any records that cannot be successfully transformed into Endpoint Catalogue FHIR R4 resources due to missing mandatory attributes or invalid source data will be excluded from the transformed dataset and recorded in a transformation failure report.

The transformation failure report will capture:

- Resource Type
- Resource Identifier
- Failure Reason

This report provides traceability for excluded records and supports remediation activities without affecting the approved migration dataset. The report will be retained alongside the transformation manifest above.

2.2 Audit Artefacts

The following artefacts will be generated during transformation:

transformation-manifest.json

transformation-failures.json

3 Validation Against targets.json

Following transformation, the migrated dataset will be validated against the existing `targets.json` file. The purpose of this validation is not to generate the migrated data but to confirm that the migrated routing configuration accurately reproduces the routing behaviour currently used in production.

The `targets.json` file represents the authoritative source of service-to-endpoint routing. Each service identifier contained within the file is associated with an endpoint URL that represents the expected routing destination.

For every service identifier present within `targets.json`, the validation process will locate the corresponding HealthcareService within the transformed FHIR dataset. The process will then follow the HealthcareService relationship to its Endpoint and Endpoint Template in order to resolve the endpoint address associated with that service.

The resolved endpoint address from the transformed dataset will be compared against the expected URL defined in `targets.json`. Comparison logic will be tolerant of differences in casing and trailing slashes to avoid false validation failures caused by formatting inconsistencies.

The validation process provides assurance that the migrated Endpoint Catalogue accurately reproduces the routing behaviour currently defined in `targets.json`.

4 Validation Outcomes

Following validation, each service will be classified into one of three categories.

4.1 Matched

A service is classified as matched when:

- A HealthcareService exists.
- The HealthcareService references a valid Endpoint.
- The Endpoint references a valid Endpoint Template.
- The resolved endpoint URL matches the URL defined within `targets.json`.

Matched services represent fully compliant and validated routing records. These services form the approved migration dataset and are eligible for promotion to the PROD environment.

4.2 Missing

A service is classified as missing when one or more required resources cannot be located. Examples include missing HealthcareServices, missing Endpoints, or missing Endpoint Templates.

These records cannot be validated and therefore will not be included in the approved migration dataset until the missing information has been resolved.

4.3 Mismatched

A service is classified as mismatched when all required resources exist, but the resolved endpoint URL does not match the URL defined in `targets.json`.

These records indicate a routing discrepancy between the migrated dataset and the current operational routing configuration. Such records must be investigated and corrected before they can be considered for migration to PROD.

5 Production Migration Approach

Once transformation and validation activities have successfully completed within INT, the resulting dataset becomes the approved migration package for production deployment.

Only resources with a validation status of **Matched** will be included in the migration package. Resources that are classified as Missing or Mismatched will be excluded from deployment and tracked separately for remediation.

The approved migration dataset will contain the complete set of validated FHIR R4 resources, including Endpoint Templates, Endpoints, and HealthcareServices. This dataset will be promoted to PROD using a controlled migration process and loaded directly into the production Endpoint Catalogue data store.

Following deployment, a lightweight verification exercise will be performed to confirm record counts, resource relationships, and successful data loading. This ensures the validated INT dataset has been accurately promoted to PROD without modification.

This approach provides a clear separation between migration, validation, and deployment activities while minimising risk and ensuring the Endpoint Catalogue is populated only with validated FHIR R4 compliant resources that accurately reflect the routing behaviour defined within `targets.json`.